# Testing and Calibrating the Section Phase Shifters and Attenuators

## CLS TECHNICAL PROCEDURE – 1.7.39.1 Rev. 0

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## REVISION HISTORY

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| A | 2001-04-25 | Original Draft | R. Mark Silzer |
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## Table of Contents

# 1. Purpose and Scope

This document describes the procedure to test and calibrate the stepper-motor-driven phase shifters and attenuators used in the linac section feed arms. This document is limited to the phase shifters and attenuators located upstairs in the modulator room (room 1032) and does not cover the phase shifters and attenuators located in the linac vault--servicing the Energy Compression System, prebuncher, chopper, or the high power phase shifter located between section 0 and section 1.

# 2. Background

The section attenuators and phase shifters are mounted from the roof of room 1032. (Refer to CLS RF System PFD of the linac, ACCL/PFD/RF/0057151 for a schematic diagram of the RF plumbing in room 1032). These units are used to control the RF drive to the linac klystrons. The RF Source provides pulses of 2856 MHz RF to a waveguide power splitter located just above the RF Source. From here, separate RF feeds are sent to a location near klystrons 1 & 2, and 3 & 4 via coaxial lines (The system to feed klystrons 5 & 6 is located near klystrons 3 & 4). Each RF feed is further split by a waveguide hybrid (Magic T). The transformation from coax to WR284 waveguide is accomplished by using a small 90-degree section of 7/8" rigid coax (ANDREW 1060A) followed by a coax-to-waveguide adapter. Each of the two forward power ports on the Magic T are followed by a waveguide phase shifter, a waveguide attenuator and a waveguide-to-coax adapter (N connector) to service each klystron. The phase shifters and attenuators are stepper-motor driven and utilize potentiometers to relay their position. Each unit is fitted with both CW and CCW limit switches.

# 3. Definitions and Abbreviations

The attenuators referred to in this document are named ATN1032-01, ATN1032-02, ATN1032-03, ATN1032-04, ATN1032-05, and ATN1032-06. The 1032 denotes the room number of their location and the 01 to 06 denotes which klystron the attenuator is servicing.

The phase shifters referred to in this document are named PHS1032-01, PHS1032-02, PHS1032-03, PHS1032-04, PHS1032-05, and PHS1032-06. The 1032 denotes the room number of their location and the 01 to 06 denotes which klystron the phase shifter is servicing.

# 4. Required Equipment

- Network analyser (example HP 8753ES).
- Two lengths of good quality RF cable (minimum length 3 meters each of RG-8 for example) preferably with male N-connector ends.
- One 7/8 inch EIA flange to female N connector adapter (example Andrew 2260B adapter)
- Pair of 7/16 inch wrenches
- Access to a computer running the dm2k (operator interface for EPICS) application on LINUX. You will need to run the CLS generic motor control software.

# 5. Procedure Steps

## 5.1 Attenuator Testing

### 5.1.1 Setup

1. Make sure that no RF is coming from the RF Drive in room 1032. Do this by turning the RF key to the off position and removing the key. As an added precaution, the RF switch on the Drive can be toggled to the OFF (down) position. RF light should not be lit.
2. At the attenuator location, disconnect the coaxial feed from the RF Drive just prior to the 90 degree mitred elbow (ANDREW 1060A) and at the N connector of waveguide-to-coaxial line (N connector) adapter.
3. Connect the N connector to 7/8 inch EIA flange adapter (Andrew 2260B adapter or equivalent) to the mitred elbow.
4. THE NEWORK ANALYSER MUST AT ALL TIMES BE KEPT WELL AWAY FROM THE PERMANENT MAGNETS THAT ARE PART OF THE KLYSTRONS. Locate the network analyser near the measurement location making sure to keep it well away from the permanent magnets located at the klystrons. An extension cord may be required to power the unit.
5. Connect the two lengths of coaxial cable to the network analyser ports. For these connections, an N adapter should be used on the ports of the network analyser. These adapters are used to save the port terminals from excessive ware caused by repetitive reconnections.
6. Connect the line from port 1 on the network analyser to the N connector of the N-to-7/8" EIA flange adapter and port 2 to the waveguide-to-N adapter.
7. The network analyser should be configured to run a S21 measurement at a frequency of 2856 MHz. Remember to perform an S21 calibration to calibrate out the effect of the cables.

### 5.1.2 Computer Setup

1. You will need access to a computer running the dm2k (operator interface for EPICS) application on LINUX and login as control. This may be one of the terminals in the Control Room (room 1021). If required, the applications can be run on a laptop.
2. To access the motor control for the attenuator and the feedback status you must run the linac or runlinac program. This will bring up the main page containing links to sections 1 through 6. Clicking on the required section will bring up the section page. Clicking on the diagnostic icon will bring up a window for running the motors and will contain information about the current calibration.

### 5.1.3 Attenuator Calibration or Calibration Check

1. To avoid setting up a standing wave in the system, you must make sure that the attenuator from the complementary section is run to its CCW limit. This is the position of maximum attenuation. The complementary sections are 1 & 2, 3 & 4, and 5 & 6.
2. To ensure the reproduceabilty of future readings, the phase shifter next to the attenuator to be tested must be driven to the CCW limit.

3. Test the operation of the limit switches. Run the attenuator stepper motor to the CW limit and record the ADC reading. Run the attenuator stepper motor to the CCW limit and record the ADC reading. Repeat the above procedure making sure that the ADC readings at the CW and CCW limits are within 2 ADC values. Running of the motors can be done using the sliders or, a right click on the slider will allow you to enter exact values. You should record your results in the table located in Appendix A.

4. With the attenuator at the CCW limit, record the S21 amplitude in dB and the ADC value. This reading will be used to normalize all other readings. Use the calibration table shown in Appendix A.

5. Record the S21 amplitude for ADC values throughout the full range of the attenuator. You should do a minimum of 20 points for the calibration curve. The final reading should be taken when the CW limit is reached. To check a current calibration, you can read the attenuation on the computer. The values should be within 0.2 dB of the current calibration once the normalization is done.

## 5.2 Phase shifter testing

### 5.2.1 Setup

1. Make sure that no RF is coming from the RF Drive in room 1032. Do this by turning the RF key to the off position and removing the key. As an added precaution, the RF switch on the Drive can be toggled to the OFF (down) position. RF light should not be lit.
2. At the phase shifter location, disconnect the coaxial feed from the RF Drive just prior to the 90 degree mitred elbow (ANDREW 1060A) and at the N connector of waveguide-to-coaxial line (N connector) adapter.
3. Connect the N connector to 7/8 inch EIA flange adapter (Andrew 2260B adapter or equivalent) to the mitred elbow.
4. THE NEWORK ANALYSER MUST AT ALL TIMES BE KEPT WELL AWAY FROM THE PERMANENT MAGNETS THAT ARE PART OF THE KLYSTRONS. Locate the network analyser near the measurement location making sure to keep it well away from the permanent magnets located at the klystrons. An extension cord may be required to power the unit.
5. Connect the two lengths of coaxial cable to the network analyser ports. For these connections, an N adapter should be used on the ports of the network analyser. These adapters are used to save the port terminals from excessive ware caused repetitive reconnections.
6. Connect the line from port 1 on the network analyser to the N connector of the N-to-7/8" EIA flange adapter and port 2 to the waveguide-to-N adapter.
7. The network analyser should be configured to run a S21 measurement at a frequency of 2856 MHz. Remember to perform an S21 calibration to calibrate out the effect of the cables.

### 5.2.2 Computer Setup

1. You will need access to a computer running the dm2k (operator interface for EPICS) application on LINUX and login as control. This may be one of the terminals in the Control Room (room 1021). If required, the applications can be run on a laptop.
2. To access the motor control for the phase shifter and the feedback status you must run the linac or runlinac program. This will bring up the main page containing links to sections 1 through 6. Clicking on the required section will bring up the section page. Clicking on the diagnostic icon will bring up a window for running the motors and will contain information about the current calibration.

### 5.2.3 Phase Shifter Calibration or Calibration Check

1. To avoid setting up a standing wave in the system, you must make sure that the attenuator from the complementary section is run to its CCW limit. This is the position of maximum attenuation. The complementary sections are 1 & 2, 3 & 4, and 5 & 6.
2. Test the operation of the limit switches. Run the phase shifter stepper motor to the CW limit and record the ADC reading. Run the phase shifter stepper motor to the CCW limit and record the ADC reading. Repeat the above procedure making sure that the ADC readings at the CW and CCW limits are within 2 ADC values. Running of the motors can be done using the sliders or, a right click on the slider will allow you to enter exact values. You should record your results in the table located in Appendix B.
3. With the phase shifter at the CCW limit, record the S21 phase in degrees and the ADC value. This reading will be used to normalize all other readings. Use the calibration table shown in Appendix B.

4. Record the S21 phase for ADC values throughout the full range of the phase shifter. You should do a minimum of 20 points for the calibration curve. The final reading should be taken when the CW limit is reached. To check a current calibration, you can read the phase on the computer. The values should be within 1 degree of the current calibration once the normalization is done.

## Appendix A- Test Sheets for Section Attenuators

ATTENUATOR TESTED ________________________

Complementary Attenuator ________________________ at CW limit ______
Check

Phase Shifter ________________________ at CCW limit ______
Check

Network Analyser S21 cable calibration done ______
Check

Limit Check

| LIMIT | LIMIT INDICATION OK (check) | ADC VALUE Trial 1 | ADC VALUE Trial 2 |
|---|---|---|---|
| CW | | | |
| CCW | | | |

Date: _______________ Tested by: ____________________ Signature: _________________

(YYYY-MM-DD) (print name)

CALIBRATION of ATTENUATOR ____________________

| POSITION | ADC VALUE | ATTENUATION READING (dB) | ATTENUATION RELATIVE TO CCW LIMIT (dB) |
|---|---|---|---|
| CCW | | | 0 |
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |
| 6 | | | |
| 7 | | | |
| 8 | | | |
| 9 | | | |
| 10 | | | |
| 11 | | | |
| 12 | | | |
| 13 | | | |
| 14 | | | |
| 15 | | | |
| 16 | | | |
| 17 | | | |
| 18 | | | |
| 19 | | | |
| 20 | | | |
| 21 | | | |
| 22 | | | |
| 23 | | | |
| 24 | | | |
| 25 | | | |
| 26 | | | |
| 27 | | | |
| 28 | | | |
| CW limit | | | |

Date: ______________ (YYYY-MM-DD) Tested by: ___________________ (print name) Signature: ________________

## Appendix B- Test Sheets for Section Phase Shifters

PHASE SHIFTER TESTED ________________________

Complementary Attenuator ________________________ at CW limit ______
Check

Network Analyser S21 cable calibration done ______
Check

Limit Check

| LIMIT | LIMIT INDICATIO N OK (check) | ADC VALUE Trial 1 | ADC VALUE Trial 2 |
|---|---|---|---|
| CW | | | |
| CCW | | | |

Date: _______________ Tested by: ____________________ Signature: _________________

(YYYY-MM-DD) (print name)

CALIBRATION of PHASE SHIFTER ______________________

| POSITION | ADC VALUE | PHASE READING (degrees) | PHASE RELATIVE TO CCW LIMIT (degrees) |
|---|---|---|---|
| CCW | | | 0 |
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |
| 6 | | | |
| 7 | | | |
| 8 | | | |
| 9 | | | |
| 10 | | | |
| 11 | | | |
| 12 | | | |
| 13 | | | |
| 14 | | | |
| 15 | | | |
| 16 | | | |
| 17 | | | |
| 18 | | | |
| 19 | | | |
| 20 | | | |
| 21 | | | |
| 22 | | | |
| 23 | | | |
| 24 | | | |
| 25 | | | |
| 26 | | | |
| 27 | | | |
| 28 | | | |
| CW limit | | | |

Date: ________________ Tested by: ______________________ Signature: __________________

(YYYY-MM-DD) (print name)